Alert Triage Workflow: Show how a security engineer might investigate and resolve a cloud security alert

Problem statement:

Cloud security systems generate multiple alerts when abnormal network traffic or unusual resource usage is detected. However, security engineers may struggle to prioritize these alerts, identify critical threats, and investigate suspicious activities efficiently. Without a clear and organized alert management interface, important threats may be overlooked or take longer to resolve.

The goal is to design a user-friendly security dashboard that organizes alerts by priority (High, Medium, and Low), provides relevant information for investigation, and helps security engineers take appropriate actions and track incident resolution. This will improve alert visibility, simplify the investigation process, and support faster, more efficient security incident management.

Proposed Approach:

1.Overview
Develop a Security Alert Monitoring and Investigation Dashboard to monitor virtual machine (VM) resource utilization and network traffic anomalies, helping security engineers identify and investigate potential threats.

2.Alert Prioritization
Classify alerts into four severity levels—Low, Medium, High, and Critical-based on resource utilization, duration of abnormal activity, network traffic deviations, and available threat indicators.

3.Key Dashboard Features
The dashboard will provide an alert overview, severity-based filtering, detailed alert information, and visual analytics for resource and network traffic trends. It will also support incident tracking through New, Investigating, Resolved, and False Positive statuses, with explanations for each alert's priority.

4.Evaluation
Evaluate the dashboard through task-based usability testing, measuring task completion time, success rate, and user feedback using the System Usability Scale (SUS).

5.Objective
The main objective is to improve alert visibility, simplify threat investigation, prioritize security incidents effectively, and enhance overall incident management efficiency.

The project aims to enhance the existing SIEM framework by integrating cloud monitoring data and improving alert prioritization. This will help security engineers identify critical threats, simplify investigations, and improve overall security incident response efficiency.